

Absolutely. Here are **two more complex examples** designed for DS exam practice. I'll show **Infix** → **Postfix** and **Infix** → **Prefix**, with stack tables and the reasoning behind each conversion.

Example 1 — Multiple operators and nested parentheses

Infix expression

A + (B * C - (D / E ^ F) * G) * H

This is useful because it contains:

- nested parentheses
- +, -, *, /, ^
- exponentiation
- different precedence levels

A. Infix → Postfix

Precedence

^ Highest
* /
+ - Lowest

^ is right-associative; the others are left-associative.

Stack conversion

Step	Symbol	Action	Stack	Output
1	A	Operand → output	—	A
2	+	Push	+	A
3	(	Push	+ (	A
4	B	Operand	+ (	AB
5	*	Push	+ (*	AB

Step	Symbol	Action	Stack	Output
6	C	Operand	+ (*	ABC
7	-	Pop *, push -	+ (-	ABC*
8	(	Push	+ (- (	ABC*
9	D	Operand	+ (- (	ABC*D
10	/	Push	+ (- (/	ABC*D
11	E	Operand	+ (- (/	ABC*DE
12	^	Push	+ (- (/ ^	ABC*DE
13	F	Operand	+ (- (/ ^	ABC*DEF
14	)	Pop ^, / ; discard (	+ (-	ABC*DEF^/
15	*	Push	+ (- *	ABC*DEF^/
16	G	Operand	+ (- *	ABC*DEF^/G
17	)	Pop *, - ; discard (	+	ABC*DEF^/G*-
18	*	Push	+ *	ABC*DEF^/G*-
19	H	Operand	+ *	ABC*DEF^/G*-H
20	End	Pop *	+	ABC*DEF^/G*-H*
21	End	Pop +	—	ABC*DEF^/G*-H*+

Postfix

ABC*DEF^/G*-H*+

With spaces:

A B C * D E F ^ / G * - H * +

B. Infix → Prefix

First understand the expression structure:

$$A + (B * C - (D / E ^ F) * G) * H$$

The outermost operator is:

+

Left operand:

A

Right operand:

$$(B * C - (D / E ^ F) * G) * H$$

The prefix representation is therefore:

$$+ A * [B * C - (D / E ^ F) * G] H$$

Now convert each part.

$$B * C$$

$$* B C$$

$$D / E ^ F$$

Because $^$ has higher precedence:

$$D / (E ^ F)$$

Prefix:

$$/ D ^ E F$$

$$(D / E ^ F) * G$$

Prefix:

$$* / D ^ E F G$$

$$B * C - (D / E ^ F) * G$$

Prefix:

$$- * B C * / D ^ E F G$$

Multiply by H

$$* - * B C * / D ^ E F G H$$

Finally add A

$$+ A * - * B C * / D ^ E F G H$$

Therefore:

Prefix

$$+ A * - * B C * / D ^ E F G H$$

or, preferably with spaces:

$$+ A * - * B C * / D ^ E F G H$$

Example 2 — More complex expression

Consider:

$$(A + B) * C - D / (E - F * G) + H ^ (I - J * K)$$

This is a very good examination example because it contains:

- multiple parentheses
- nested operations
- $+$, $-$, $*$, $/$, $^$
- exponentiation
- several subexpressions

A. Infix → Postfix

Expression:

$(A+B) * C - D / (E - F * G) + H ^ (I - J * K)$

Step-by-step

Step	Symbol	Action	Stack	Output
1	(	Push	(	
2	A	Output	(	A
3	+	Push	(+	A
4	B	Output	(+	AB
5	)	Pop +	—	AB+
6	*	Push	*	AB+
7	C	Output	*	AB+C
8	-	Pop *, push -	-	AB+C*
9	D	Output	-	AB+C*D
10	/	Push	- /	AB+C*D
11	(	Push	- / (	AB+C*D
12	E	Output	- / (	AB+C*DE
13	-	Push	- / (-	AB+C*DE
14	F	Output	- / (-	AB+C*DEF
15	*	Push	- / (- *	AB+C*DEF
16	G	Output	- / (- *	AB+C*DEFG
17	)	Pop *, -	- /	AB+C*DEFG*-
18	+	Pop / , then - ; push +	+	AB+C*DEFG*- / -

Let's write step 18 carefully:

Before + :

Stack = [-, /]

Since / has higher precedence:

Pop /

Then - has equal precedence with + and is left-associative:

Pop -

Output becomes:

AB+C*DEFG*-/-

Then push + .

Step	Symbol	Action	Stack	Output
19	H	Output	+	AB+C*DEFG*-/H
20	^	Push	+ ^	AB+C*DEFG*-/H
21	(	Push	+ ^ (	AB+C*DEFG*-/H
22	I	Output	+ ^ (	AB+C*DEFG*-/HI
23	-	Push	+ ^ (-	AB+C*DEFG*-/HI
24	J	Output	+ ^ (-	AB+C*DEFG*-/HIJ
25	*	Push	+ ^ (- *	AB+C*DEFG*-/HIJ
26	K	Output	+ ^ (- *	AB+C*DEFG*-/HIJK
27	)	Pop *, - ; discard (	+ ^	AB+C*DEFG*-/HIJK*-
28	End	Pop ^	+	AB+C*DEFG*-/HIJK*-^
29	End	Pop +	—	AB+C*DEFG*-/HIJK*-^+

Final Postfix

$AB+C*DEFG*-/HIJK*-\wedge+$

With spaces:

$A\ B\ +\ C\ *\ D\ E\ F\ G\ *\ -\ /\ -\ H\ I\ J\ K\ *\ -\ \wedge\ +$

B. Infix $\rightarrow$ Prefix

Let's derive it from the expression tree.

Original:

$(A+B)*C - D/(E-F*G) + H^\wedge(I-J*K)$

Because $+$ and $-$ have the same precedence and are left-associative:

$((A+B)*C - D/(E-F*G)) + H^\wedge(I-J*K)$

Therefore the outermost operator is:

$+$

Left side

$(A+B)*C - D/(E-F*G)$

The main operator is $-$.

First part

$(A+B)*C$

Prefix:

$*\ +\ A\ B\ C$

Second part

$$D / (E - F * G)$$

Inside:

$$E - F * G$$

Since $*$ has higher precedence:

$$E - (F * G)$$

Prefix:

$$- E * F G$$

Therefore:

$$D / (E - F * G)$$

becomes:

$$/ D - E * F G$$

So the entire left side becomes:

$$- * + A B C / D - E * F G$$

Right side

$$H^{\wedge} (I - J * K)$$

Inside:

$$I - J * K$$

becomes:

$$I - (J * K)$$

Prefix:

- I * J K

Therefore:

$H^{\wedge}(I-J*K)$

becomes:

$^{\wedge} H - I * J K$

Combine both sides

Outer operator:

+

Therefore:

$+ - * + A B C / D - E * F G$
 $^{\wedge} H - I * J K$

Final Prefix

$+ - * + A B C / D E * F G ^{\wedge} H - I * J K$

With spaces:

$+ - * + A B C / D - E * F G ^{\wedge} H - I * J K$

3. Final Answers — Both Examples

Example	Infix	Postfix	Prefix
1	$A + (B * C - (D / E ^{\wedge} F) * G) * H$	$A B C * D E F ^{\wedge} / G * - H * +$	$+ A * - * B C * / D ^{\wedge} E F G H$
2	$(A + B) * C - D / (E - F * G) + H ^{\wedge} (I - J * K)$	$A B + C * D E F G * - / H I J K * -$ $^{\wedge} +$	$+ - * + A B C / D E * F G ^{\wedge} H - I * J K$

Important patterns to remember

Example 1

$A + (B * C - (D / E ^ F) * G) * H$

Postfix:

$A B C * D E F ^ / G * - H * +$

Prefix:

$+ A * - * B C * / D ^ E F G H$

Example 2

$(A + B) * C - D / (E - F * G) + H ^ (I - J * K)$

Postfix:

$A B + C * D E F G * - / - H I J K * - ^ +$

Prefix:

$+ - * + A B C / D - E * F G ^ H - I * J K$

These two examples cover most of the **stack-based infix** → **postfix/prefix conversion cases** you are likely to encounter in a Data Structures exam.